

MATTHEW CAMP

Huntsville, AL ■ 423.322.3976 ■ resume@aicampground.com

PROFILE

Senior Cloud Infrastructure Engineering Leader with 15+ years of technical leadership experience and deep expertise in Kubernetes orchestration, container platforms, and cloud-native architecture. Proven track record leading cross-functional engineering teams while driving infrastructure modernization initiatives across government and private sectors. Expert in multi-cluster deployments, service mesh technologies, and GitOps practices with extensive open-source contributions spanning 8+ years.

- Demonstrated 15+ years of leadership experience managing technical teams and driving infrastructure transformation initiatives from Marine Corps operations through senior engineering roles.
- Deep technical expertise in Kubernetes cluster management, containerization, and cloud-native platform development with hands-on experience across AWS, multi-cluster environments, and service mesh implementation.
- Extensive open-source contributor with 8+ years of contributions to Neovim plugins, Nix ecosystem projects, and infrastructure automation tools.
- Proven success in developing and implementing comprehensive DevOps strategies including CI/CD optimization, Infrastructure as Code, and GitOps practices.
- Current Top Secret/Specialized Compartmentalized Information (TS/SCI) with Counterintelligence Polygraph.

CORE COMPETENCIES

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|-----------------------------|---------------------------------|----------------------------|
| • K8s Cluster Management | • Container Orchestration | • Observability/Monitoring |
| • Cloud Infra Leadership | • GitOps/Infrastructure as Code | • Open Source Development |
| • Service Mesh Technologies | • Technical Team Leadership | • Multi-Cloud Architecture |

PROFESSIONAL EXPERIENCE

RELEVANT EXPERIENCE

Advanced Technology Applications (ATA) LLC - Great Falls, VA (Remote)

Senior Data Scientist / Full Stack MLOps/DevOps Engineer

November 2021 - Present

Senior technical contributor on the platform team with responsibility for delivering production-grade AI/ML capabilities alongside secure, reproducible infrastructure. Led applied machine learning efforts spanning dataset creation, model development, and deployment while shaping team-wide MLOps and DevSecOps practices. Combined expertise in data science, distributed systems, and platform engineering to deliver mission-ready analytics solutions for government customers and mentor junior engineers.

Key Accomplishments

- Stood up and administered Label Studio to create high-quality labeled training datasets for computer vision models including annotating satellite imagery to identify maritime vessels and other domain-specific features.
- Developed and trained machine learning models for image segmentation and classification including a UNet-based computer vision model in Julia with experiment tracking and evaluation using MLFlow.
- Designed and implemented reproducible MLOps workflows supporting model development and deployment across local and Kubernetes environments enabling consistent and auditable ML experimentation.
- Spearheaded adoption of Nix and NixOS to establish a secure deterministic DevSecOps and MLOps framework eliminating configuration drift and ensuring consistent ML runtimes across teams.
- Streamlined deployment of ML-enabled Flink jobs using Nix enabling consistent execution across developer workstations and production Kubernetes clusters.
- Built a single-command local Kubernetes development environment using K3D and Nix significantly reducing setup time for data science and ML experimentation.
- Deployed and customized JupyterHub on enterprise Kubernetes clusters providing integrated Python Julia and R environments for applied data science and machine learning workflows.

- Designed modular scalable infrastructure for data-intensive systems including Kafka and MLFlow improving reliability observability and security of ML workloads.
- Mentored junior developers and data scientists in applied machine learning MLOps best practices and reproducible development environments accelerating onboarding and improving team effectiveness.
- Orchestrated a complete CI/CD pipeline for a large-scale medical form processing system using AWS Textract and Lambda functions leveraging Nix to enable fully reproducible air-gapped deployments of the entire software and ML stack.

Colsa - Huntsville, AL

Artificial Intelligence Software Developer / Full Stack Data Scientist / Tech Lead September 2018 - November 2021

Technical leader driving DevOps transformation and infrastructure modernization in secure government environments. Led implementation of containerized deployment strategies, CI/CD pipeline optimization, and monitoring frameworks on air-gapped systems. Mentored development teams in modern software engineering practices while developing data science infrastructure.

Key Accomplishments

- Led organizational transformation implementing GitLab CI/CD pipelines Docker containerization and GitOps practices establishing automated testing and deployment workflows.
- Created standardized DevOps practices through cookiecutter project templates and pipeline automation empowering development teams with consistent deployment capabilities.
- Implemented comprehensive observability and monitoring solutions using Grafana and Prometheus for distributed application performance tracking.
- Architected and developed production data pipeline infrastructure using Docker and Apache Airflow demonstrating advanced container orchestration and workflow management.
- Built production-ready GraphQL API infrastructure in Python with automated deployment and configuration management practices.
- Designed and implemented domain-specific graph database infrastructure (MongoDB + Neo4j) with automated deployment pipelines.

Science Applications International Corporation (SAIC) - Huntsville, AL

Senior Database Administrator & Business Process Improvement Analyst August 2017 - August 2018

Technical infrastructure leader responsible for automation strategy and cloud application development within security department. Led technical initiatives for enterprise system integration and process automation while mentoring junior technical staff in modern development practices.

Key Accomplishments

- Led technical team initiatives developing real-time automated reporting infrastructure using Flask web application architecture integrated with SQL Server and enterprise systems.
- Implemented automated enterprise integration solutions using Python Selenium and AutoIT reducing manual processes and improving operational efficiency.
- Developed autonomous system processing capabilities handling hundreds of weekly transactions with high reliability and automated workflow management.
- Implemented Lean project management practices using self-hosted Kanban systems demonstrating process improvement and technical workflow optimization.

ExoAnalytic Solutions - Huntsville, AL

Algorithm Developer April 2016 - August 2017

Technical infrastructure engineer supporting distributed computing environments and high-performance system administration. Managed large-scale data processing systems while providing comprehensive IT infrastructure leadership for cloud computing initiatives.

Key Accomplishments

- Architected and implemented deep learning infrastructure using Python and TensorFlow for processing multi-million record datasets in cloud computing environments.
- Led database modernization project converting complex MATLAB data structures to optimized SQLite implementations improving system performance and scalability.
- Implemented graph database infrastructure using Neo4j for Customer Relationship Management system demonstrating modern database architecture expertise.
- Developed automated data pipeline solutions using Python for business intelligence and operational monitoring capabilities.

Elbit Systems of America - Huntsville, AL

Marketing Intern

September 2015 - April 2016

Supported technical architecture planning and system design initiatives for enterprise Customer Relationship Management infrastructure.

Key Accomplishments

- Collaborated with cross-functional teams to architect database schema and system requirements for enterprise CRM infrastructure.
- Conducted technical research and analysis supporting strategic planning initiatives and system architecture design.

Intelligence Support Battalion, FHG, Marine Forces Reserve USMC - New Orleans, LA

Training & Operations Supervisor/Intelligence Analyst

March 2012 - February 2015

Led technical teams and infrastructure development across 28 geographically dispersed locations. Managed cross-functional engineering initiatives while developing automated systems for real-time operational visibility and distributed team collaboration.

Key Accomplishments

- Led distributed technical teams across 28 locations through development of integrated SharePoint and database systems providing real-time operational visibility.
- Developed and implemented automated systems using Visual Basic for Applications (VBA) with seamless integration to Marine Corps enterprise training systems enabling real-time operational visibility.
- Created comprehensive technical documentation and training materials for deployed systems ensuring knowledge transfer and sustainable operations.
- Awarded Navy and Marine Corps Commendation Medal for technical leadership and distributed system implementation achievements.

United States Marine Corps Reserve - Chattanooga, TN

Artillery Section Chief

2004 - 2010 - 2011 - 2012

Led technical teams of 15 Marines in complex equipment operations and system management. Developed comprehensive training programs while maintaining operational excellence in high-pressure environments including combat deployments.

EDUCATION

Artificial Intelligence Nanodegree – February 2018
Udacity

Deep Learning Nanodegree – May 2017
Udacity

Master of Science in Management Science - Business Analytics – May 2017
University of Alabama in Huntsville – Huntsville, AL

Bachelor of Arts in History – May 2016
University of Alabama in Huntsville – Huntsville, AL

TECHNICAL SKILLS

Kubernetes (K8s) ■ Docker/Containerization ■ Istio/Service Mesh ■ Grafana/Prometheus ■ GitOps/ArgoCD ■
Terraform/IaC ■ AWS/Multi-Cloud ■ Python/Scripting ■ GitLab CI/CD ■ Linux Systems ■ Nix/NixOS ■ Apache Airflow ■
JavaScript/Lua/Rust ■ MongoDB/Neo4j ■ Flask/API Development